



Jhankar Bansal

B.Tech — Computer Science and Engineering — 2023–2027

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Portfolio: Portfolio — Engineering Playbook — Change Blueprint

Education

Bachelor of Technology (B.Tech), Computer Science and Engineering

2023–2027

CGPA: 8.04

Higher Secondary Education (Science)

Central Board of Secondary Education

Percentage: 70%

Secondary Education

Central Board of Secondary Education

Percentage: 95.35%

Skills

Programming Languages: Python, C++, JavaScript

Frameworks & Libraries: React, Node.js, Express

Databases & Tools: MongoDB, Git, GitHub, Linux

Concepts: OOPs, Backend, Theory of Computation, Database Management System

Projects

PrepGrounded – Evidence-Grounded Placement Preparation Engine

Jan 2026 – Present

Tech: React, Node.js, Express, JavaScript, Vector Embeddings, Cheerio, Node-Cron, Git

- Engineered a Time-Aware RAG Retrieval Engine combining Cosine Vector Similarity (60%) with Exponential Temporal Recency Decay (40%, $\lambda = 0.35$), achieving a +191% increase in Freshness@10 and –100% reduction in stale (>4y) experiences across a 1,020+ record corpus covering 33 tech companies.
- Architected an Evidence Quality V2 Audit Engine evaluating 6 structural dimensions (source provenance, verbatim topic grounding, round detail, outcome integrity) with hard quality caps, emitting diagnostic flags (unsupported_topics, unsupported_outcome) and clean evidence badges.
- Implemented an Automated 7-Day Live Ingestion Pipeline using Node.js, Cheerio, and Node-Cron featuring 2-stage idempotent deduplication (URL normalization + SHA-256 content hashing), schema quality gates (staging.json), and instant vector cache warmup.
- Designed a Modern SaaS Product UX & Retrieval Research Lab in React and Vite featuring human-readable result explanations ("Why recommended?"), progressive company drift insights, placement readiness skill gap reports, and a 3-arm model benchmark.

Research

BML Munjal University AI-Driven Drug Repurposing

January 2026 - Present

- Developing a Platform that combine molecular graph learning, protein–ligand interaction modeling, generative chemistry, and clinical trial data.
- Identifying new therapeutic uses for existing drugs

Achievements & Responsibilities

- TechSparX.I 2025 – KharchaCheck just won 1st place among 250+ projects at TechSparX.I 2025, hosted by the Institute of Innovation And Entrepreneurship (I2E).

- Social Media Member 2025 -Managed and curated social media accounts for the EIS 3.0 Conference,